**P2.py save model**

```
# import lib
import pandas as pd
from nltk import word_tokenize
from nltk.corpus import stopwords
from string import  punctuation
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import  MultinomialNB
from pickle import  dump

# load the data
data = pd.read_csv("movie_review_2.csv")
print(data)

# clean the data
sw = stopwords.words("english")
def clean_review(txt):
	txt = txt.lower()
	txt = word_tokenize(txt)
	txt = [t for t in txt if t not in punctuation]
	txt = [t for t in txt if t not in sw]
	txt = " ".join(txt)
	return txt

data["clean_review"] = data["review"].apply(clean_review)
print(data)

# features and target
cv = CountVectorizer()
vector = cv.fit_transform(data["clean_review"])
features = pd.DataFrame(vector.toarray(), columns=cv.get_feature_names_out())
target = data["result"]

print(features)
print(target)

# model
model = MultinomialNB()
model.fit(features.values, target)

# model and cv save
with open("model.pkl", "wb") as f:
	dump(model, f)
	print("model saved")

with open("cv.pkl", "wb") as f:
	dump(cv, f)
	print("cv saved")
```

**P3.py use model**

```
# import lib
from nltk import word_tokenize
from nltk.corpus import stopwords
from string import  punctuation
from pickle import  load

# clean the data
sw = stopwords.words("english")
def clean_review(txt):
	txt = txt.lower()
	txt = word_tokenize(txt)
	txt = [t for t in txt if t not in punctuation]
	txt = [t for t in txt if t not in sw]
	txt = " ".join(txt)
	return txt

# model and cv back
with open("model.pkl", "rb") as f:
	model = load(f)
	print("model ready")

with open("cv.pkl", "rb") as f:
	cv = load(f)
	print("cv ready")

# prediction
re = input("enter review ")
cre = clean_review(re)
vre = cv.transform([cre])
ans = model.predict(vre)
if ans[0] == "p":
	print("wow we made u happy")
else:
	print("sorry to let u down")
```

**Web app**

**S1:**

```
C:\nlp_aug26\L4>mkdir p1_mr_app
C:\nlp_aug26\L4>cd p1_mr_app
C:\nlp_aug26\L4\p1_mr_app>notepad app.py
C:\nlp_aug26\L4\p1_mr_app>mkdir templates
C:\nlp_aug26\L4\p1_mr_app>cd templates
C:\nlp_aug26\L4\p1_mr_app\templates>notepad home.html
```

**S2:**

**app.py**

```
from  flask  import  *
from nltk import word_tokenize
from nltk.corpus import stopwords
from string import  punctuation
from pickle import  load

# clean the data
sw = stopwords.words("english")
def clean_review(txt):
	txt = txt.lower()
	txt = word_tokenize(txt)
	txt = [t for t in txt if t not in punctuation]
	txt = [t for t in txt if t not in sw]
	txt = " ".join(txt)
	return txt

# model and cv back
with open("model.pkl", "rb") as f:
	model = load(f)
	print("model ready")

with open("cv.pkl", "rb") as f:
	cv = load(f)
	print("cv ready")

app = Flask(__name__)

@app.route("/", methods=["POST", "GET"])
def home():
	if request.method == "POST":
		re = request.form.get("review")
		cre = clean_review(re)
		vre = cv.transform([cre])
		ans = model.predict(vre)
		if ans[0] == "p":
			msg = "wow we made u happy"
		else:
			msg = "sorry to let u down"
		return render_template("home.html", msg=msg)
	else:
		return render_template("home.html")

if __name__ == "__main__":
	app.run(debug=True, use_reloader=True)
```

**home.html**

```
<form method="POST">
	<textarea
		name="review"
		rows=5
		cols=30
		required
	></textarea>
	<br/><br/>
	<input type="submit"
		/>
</form>
<h2> {{ msg }} </h2>
```

**S3:**

```
C:\nlp_aug26\L4\p1_mr_app\templates>cd..
C:\nlp_aug26\L4\p1_mr_app>python app.py
```